Gene signature-guided drug screening identified narciclasine as a potential therapeutic for interstitial fibrosis of the kidney

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Abstract

Introduction: Chronic Kidney Disease (CKD) is marked by progressive tubulointerstitial fibrosis (TIF), a pathological feature insufficiently addressed by existing therapies.
Methods: To identify drugs with potential to halt TIF progression, we constructed a TIF-specific gene expression signature using published human CKD kidney transcriptome data and employed the small molecule perturbant library LINCS L1000 database for a high-throughput screening of compounds capable of reversing the expression of TIF-related genes.
Results: Narciclasine, a natural compound derived from the Narcissus (amaryllis) plant, was identified as a top compound which significantly reversed the gene expression signature of TIF. Administration of narciclasine not only significantly prevented inflammation and fibrotic lesions induced by unilateral ureteral obstruction and unilateral ischemia-reperfusion injury but also delayed the progression of established TIF induced by unilateral ureteral obstruction. Furthermore, in the 5/6 nephrectomy-induced CKD model, narciclasine significantly lowered serum creatinine, reduced proteinuria, and alleviated TIF and inflammation.
Conclusions: Mechanistically, narciclasine reversed the failed-repair phenotype of tubular epithelial cells and inhibited fibroblasts proliferation and activation, at least partially via inhibiting the activation of NF-κB signaling. Our findings suggest that narciclasine should be further investigated as a promising drug candidate to attenuate CKD.

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KEYWORDS: chronic kidney disease; gene expression signatures; high-throughput drug screening; narciclasine; renal interstitial fibrosis

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Translational Statement

We identified a gene expression signature associated with tubulointerstitial fibrosis (TIF) using transcriptome data from patients with chronic kidney disease (CKD). Leveraging the Library of Integrated Network-based Cellular Signatures L1000 database, we performed high-throughput screening and successfully identified that narciclasine could reverse the TIF gene expression signature. *In vitro* and *in vivo* experiments confirmed that narciclasine significantly delayed the progression of TIF, reduced proteinuria, and improved renal function across multiple preclinical mouse models. Our findings call for further studies to evaluate the efficacy and safety of narciclasine for treating CKD.

Chronic kidney disease (CKD) is a growing global health challenge, affecting over 10% of the population worldwide.[1–3] Besides, CKD is associated with an increased risk of cardiovascular diseases and accounts for approximately 1.5% of global mortality. Unfortunately, current therapeutic options for CKD remain limited.[4,5] Renal fibrosis is a defining feature and represents a common final pathway of virtually all kinds of CKD.[6] The severity of kidney fibrosis, especially tubulointerstitial fibrosis (TIF), correlates inversely with renal function, making TIF a critical therapeutic target in CKD.[6,7] However, despite that numerous molecular targets linked to TIF have been identified, their translation into clinical applications remains elusive. This is primarily because targeting a single pathway or molecule is unlikely to halt the progression of CKD, a disease characterized by complex, multifactorial signaling networks.[7,8]

Phenotype-based drug screening offers a systematic approach to drug discovery based on the observed therapeutic effects in disease models.[8] Advances in high-throughput

screening technologies, such as organoid microarrays and cell-based imaging platforms, have further propelled this strategy.[9,10] Of note, over half of the US Food and Drug Administration–approved clinical drugs were discovered through phenotype-based strategies.[8,11] The selection of appropriate disease phenotypes is critical to the success of phenotype-based strategies. Recently, gene expression signature (GeneSig) specific to particular diseases has gained acceptance as a viable phenotype, where disease amelioration is viewed as the reversal of disease-associated GeneSig to a more normal state.[8,12] For instance, nintedanib, an approved therapeutic drug for idiopathic pulmonary fibrosis, was identified based on the GeneSig of human idiopathic pulmonary fibrosis.[13] Similarly, using GeneSig from aging and disease in *Col4a3*$^{-/-}$ mice, vorinostat was identified as a potential drug for renal fibrosis.[14] In addition, Zhang *et al.*[15] found BI2536 to be promising for diabetic nephropathy using GeneSig from kidney tissue of diabetic patients or mice. We believe that GeneSig derived from patients' data offers greater clinical relevance. Given the critical role of TIF in the progression of CKD, we assembled transcriptomic data of patients with CKD to construct TIF GeneSig by harnessing publicly available databases. Using the Library of Integrated Network-based Cellular Signatures (LINCS) L1000 dataset database, we identified narciclasine (Ncls), a natural plant product from the Narcissus family, as a promising candidate due to its potent ability to reverse the TIF GeneSig. The antifibrotic efficacy of Ncls was further validated in several preclinical models, including unilateral ischemia-reperfusion injury (UIRI), unilateral ureteral obstruction (UUO), and 5/6 nephrectomy-induced CKD models. Collectively, these data demonstrate that Ncls is a promising drug candidate for CKD treatment.

METHODS

Collection of human renal tubulointerstitial transcriptomic data

Human renal tubulointerstitial transcriptomic data were collected from NephroSeq (nephroseq.com) and NCBI Gene Expression Omnibus, focusing on datasets labeled from microdissected "tubulointerstitial" kidney samples with healthy controls. Differential gene expression analysis was conducted to identify differentially expressed genes (DEGs) between the disease groups and healthy controls in each dataset. Genes were selected based on differential expression fold-changes (FCs) greater than 2-fold or less than minus 2-fold ($\log_2$FC >1 or <-1), and a false discovery rate <0.05, forming the GeneSig for each dataset.

Detailed methodologies are provided in the Supplementary Methods.

Drug repurposing via LINCS L1000 connectivity map analysis

LINCS L1000CDS[2] includes an interactive interface where we can input upregulated and downregulated GeneSigs. We input the GeneSig from each CKD dataset into the platform, which then matches these with drug GeneSigs, displaying those small-molecule compounds that can most effectively reverse the input "tubulointerstitial" GeneSig. The results are sorted by the percentage of reversed genes, and the database displays the top 50 small-molecule compounds.

Detailed methodologies are provided in the Supplementary Methods.

Animal experiments

All animal procedures were conducted in compliance with the ethical guidelines approved by the Animal Care Ethics Committee of Nanfang Hospital (Application No. IACUC-LAC-20230210-002). Male C57BL/6 and CD-1 mice, aged 8–10 weeks, were purchased from Yancheng Biological. The UIRI and UUO models were established using C57BL/6 mice. CD-1 mice, commonly used in previous studies for 5/6 nephrectomy models, were used in our study to develop the model.[16–18] Detailed methodologies for the animal experiments are provided in the Supplementary Methods.

Drug preparation

Ncls was purchased from MedChem Express (Catalog number: HY-16563). Ncls was supplied in vials of 5 mg each. To prepare the solution, 200 μl of dimethylsulfoxide was added to the vial, and the mixture was thoroughly vortexed until Ncls was completely dissolved, achieving a concentration of 0.025 mg/μl (81.43 mM). Before each gavage administration, the stock solution of Ncls at a concentration of 0.025 mg/μl was thawed and diluted based on the mouse's body weight (e.g., 1 μl for a 25 g mouse) up to 100 μl with saline for the gavage procedure.

Lisinopril was acquired from MedChem Express (Catalog number: HY-18206). Lisinopril is dissolved in saline according to the manufacturer's instructions. The solution was prepared at a concentration of 2.5 mg/ml and then aliquoted for storage at –80 °C. For each gavage administration, the required volume was calculated based on a dose of 10 mg/kg body weight of the mouse.

Detailed methodologies for the animal experiments are provided in the Supplementary Methods.

Statistical analysis

Statistical analysis was conducted with GraphPad Prism software (version 9.1). Results from at least 3 independent experiments were displayed as means ± SD. For variables following a normal distribution, comparisons between 2 groups were made using the unpaired 2-tailed Student *t* test. One- or two-way analysis of variance, accompanied by the Tukey or Sidak test, was used for analyzing differences across multiple groups. For data not conforming to normal distribution, the Mann-Whitney *U* test was used for comparing 2 groups. A *P* value of <0.05 was considered statistically significant; $^{\#}P < 0.05$, $^{*}P < 0.01$.

Additional and more detailed experimental methods are described in the Supplementary Methods.

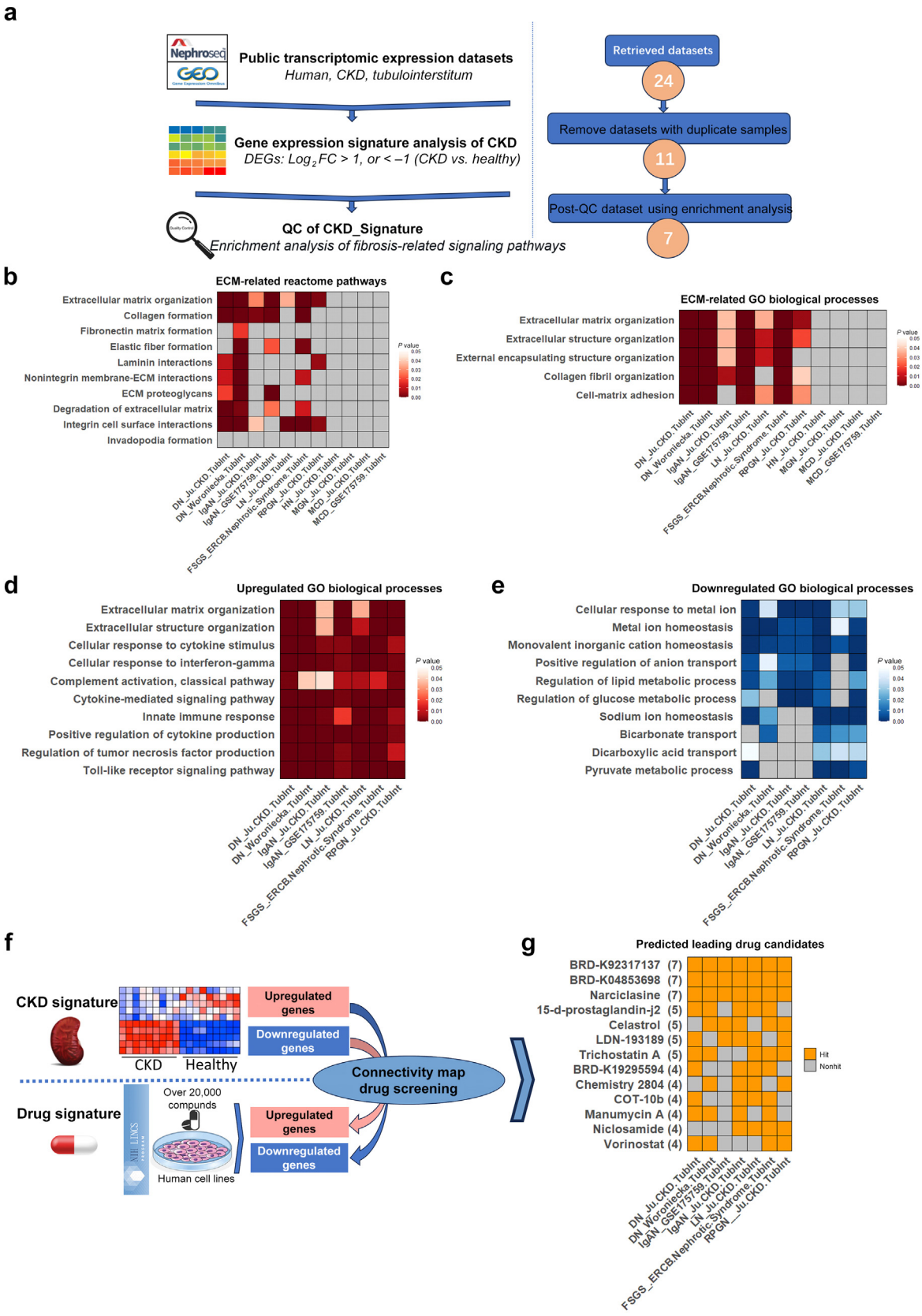


Figure 1 | Integrated analysis of renal tubulointerstitial transcriptomic data derived from patients with chronic kidney disease (CKD) and prediction of potential antifibrotic drugs. (**a**) The workflow diagram illustrating the process of gene expression signature analysis and quality control (QC). (**b**) Reactome pathway enrichment analysis conducted separately for each dataset, focusing on signaling pathways related to extracellular matrix (ECM) organization. The gray blocks indicate $P > 0.05$. (**c**) Enrichment analysis of Gene Ontology (continued)

Table 1 | Human renal tubulointerstitial transcriptome dataset for drug screening

Nephropathy	Data name	Data source	GSE ID	Sample description
DN	DN_Ju CKD TubInt	Nephroseq.com	GSE47185	17 DN and 31 healthy controls
DN	DN_Woroniecka Diabetes TubInt	Nephroseq.com	GSE30122	10 DN and 12 healthy controls
IgAN	IgAN_Ju CKD TubInt	Nephroseq.com	GSE35488	25 IgAN and 31 healthy controls
IgAN	IgAN_GSE175759	GEO	GSE175759	46 IgAN and 22 healthy controls
LN	LN_Ju CKD TubInt	Nephroseq.com	GSE32591	32 LN and 31 healthy controls
FSGS	FSGS_ERCB TubInt	Nephroseq.com	NA	18 FSGS and 9 healthy controls
RPGN	RPGN__Ju CKD TubInt	Nephroseq.com	GSE47185	21 RPGN and 31 healthy controls

CKD, chronic kidney disease; DN, diabetic nephropathy; ERCB, European Renal cDNA Bank; FSGS, focal segmental glomerulosclerosis; GEO, Gene Expression Omnibus; IgAN, IgA nephropathy; LN, lupus nephritis; MCD, minimal change disease; NA, not available; RPGN, rapidly progressive glomerulonephritis; TubInt, tubulointerstitium.

RESULTS

Collection and quality control of renal tubulointerstitial transcriptomic data in patients with CKD

We queried the NephroSeq (nephroseq.org) and Gene Expression Omnibus databases using the terms "Human," "CKD," and "Tubulointerstitium," and retrieved 24 transcriptomic datasets (Figure 1a and Supplementary Table S1). Each dataset included a healthy control group and a CKD group. The samples included in these 24 transcriptomic datasets consist of microdissected renal interstitial tissues. We excluded those datasets with substantial overlap in disease samples and those containing fewer than 6 disease samples (Supplementary Table S1). Consequently, 11 datasets were selected for the subsequent analysis (Supplementary Tables S1 and S2). Compared with healthy control groups, DEGs with FCs greater than 2 or less than –2 ($\log_2$FC >1 or <–1, false discovery rate <0.05) were considered as the GeneSig in each dataset (Supplementary Table S3).

To assess whether the GeneSig reflected the features of TIF, we performed Reactome enrichment analysis on upregulated genes in each GeneSig from the 11 datasets. The primary Reactome term for fibrosis-related pathways is "Extracellular Matrix (ECM) Organization," which encompasses 9 secondary terms related to ECM (Figure 1b). Our analysis identified that the GeneSig of 7 datasets enriched ECM-related Reactome pathways (Figure 1b). In addition, we applied Gene Ontology (GO) enrichment analysis using ECM formation-related terms and obtained consistent results (Figure 1c). Therefore, we chose the 7 datasets, which showed significant enrichment of fibrosis-related pathways, for subsequent drug screening (Table 1 and Supplementary Table S1). The datasets used for drug screening (Table 1) were previously integrated and published in 2 studies by Ju *et al.*[19] and Park *et al.*[20]

We further analyzed additional biological processes related to TIF in the 7 selected datasets. GO enrichment analysis showed that, besides ECM formation, biological processes related to inflammation and immune responses were among the most significantly activated pathways (Figure 1d), which are involved in CKD.[21,22] Conversely, downregulated genes were enriched in biological processes associated with ion homeostasis, transport, and energy metabolism, reflecting the dysfunction of renal tubules (Figure 1e).[23] Collectively, these results indicated that the extracted GeneSig represents the fundamental pathophysiological characteristics of TIF.

Prediction of potential novel drugs via LINCS L1000 Connectivity Map analysis

We used the LINCS L1000CDS[2] platform, a gene signature search engine based on Connectivity Map analysis, to predict potential therapeutic drugs (Figure 1f). The algorithm identified the top 50 compounds, out of over 22,000, that most effectively reversed the inputted GeneSig.[24] This prediction was based on the degree of overlap between the disease-associated GeneSig and the drug-associated GeneSig.[24,25] The GeneSig from each CKD dataset was inputted into the interactive interface of the platform, respectively, because we believe that screening every GeneSig from 7 datasets can be viewed as independent experimental replicates. Compounds that were identified consistently across multiple datasets were deemed more reliable. Therefore, we calculated the frequency at which each compound was identified across the 7 CKD datasets. The 13 compounds that were consistently identified in more than half of the datasets are listed in Figure 1g.

Among the 13 compounds identified, 6 have previously been reported to possess therapeutic effects on human CKD or animal models, including celastrol, vorinostat,

Figure 1 | (continued) (GO) biological processes in each dataset with a specific emphasis on terms associated with ECM organization. The gray blocks indicate $P > 0.05$. (**d**) GO biological process enrichment analysis performed on the upregulated genes within the quality-controlled dataset. The figure presents the GO terms that exhibit statistical significance across different datasets. (**e**) GO biological process enrichment analysis conducted on the downregulated genes within the quality-controlled dataset. The Y-axis is arranged according to the frequency of statistical significance for each term across different datasets. The gray blocks indicate $P > 0.05$. (**f**) Diagram of drug screening strategy based on the LINCS L1000 Connectivity Map. (**g**) Top 13 candidate drugs from screening. The drugs are sorted by the cumulative hit count in different datasets, with the hit count shown in parentheses. DEG, differentially expressed gene; DN, diabetic nephropathy; GEO, Gene Expression Omnibus; IgAN, IgA nephropathy; LN, lupus nephritis; MCD, minimal change disease; MGN, membranous glomerulonephritis; RPGN, rapidly progressive glomerulonephritis; Tubint; tubulointerstitium.

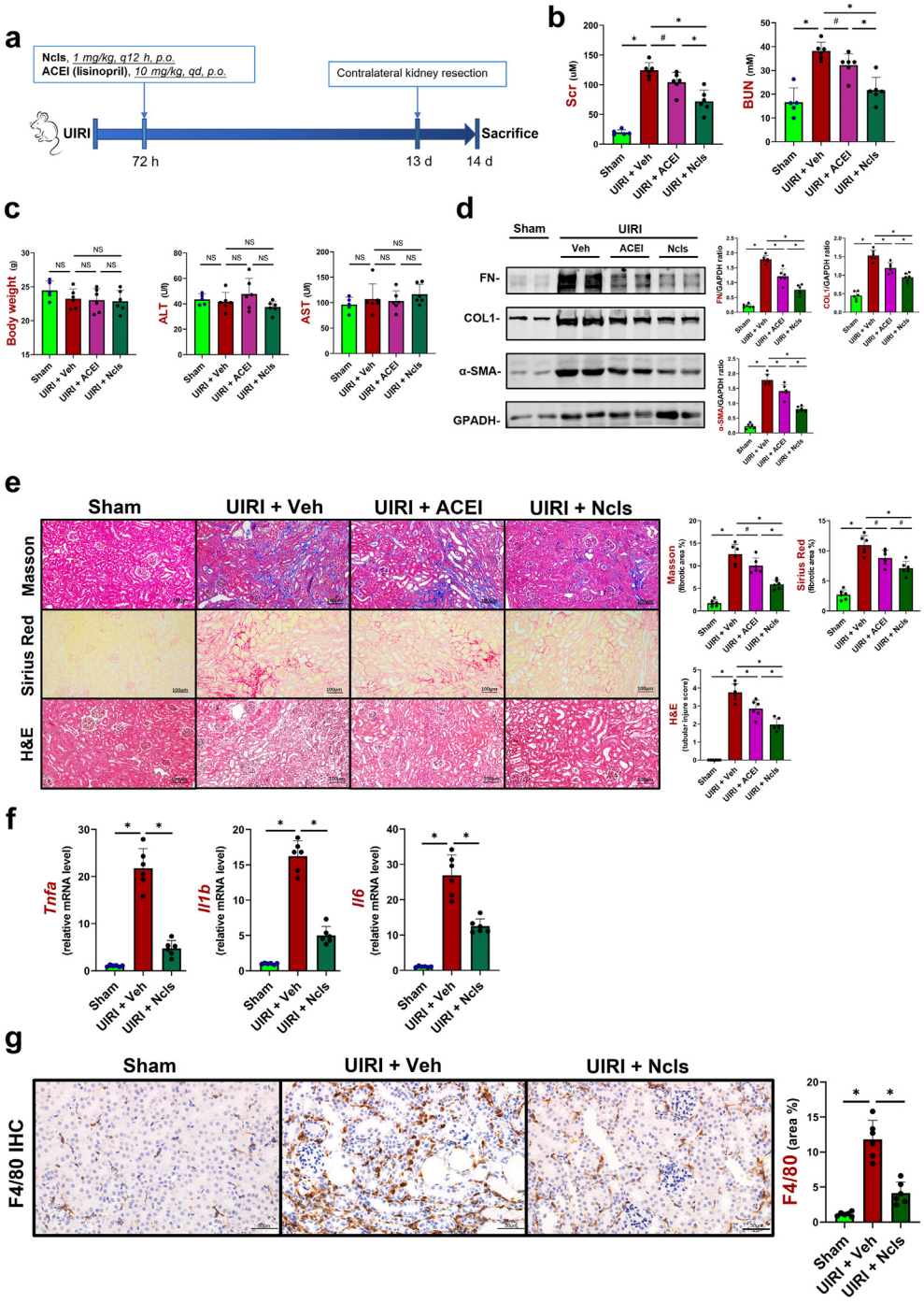


Figure 2 | Narciclasine (Ncls) alleviates tubulointerstitial fibrosis (TIF) and inflammation induced by unilateral ischemia-reperfusion injury (UIRI). (**a**) Experimental design schematic in the UIRI mouse model. (**b**) Serum creatinine (Scr) and blood urea nitrogen (BUN) levels in different groups. Sham (n = 5), UIRI + vehicle (Veh; n = 6), UIRI + angiotensin-converting enzyme inhibitor (ACEI; n = 6), and UIRI + narciclasine (Ncls; n = 6). (**c**) Body weight, aspartate aminotransferase (AST), and alanine aminotransferase (ALT) levels in different groups. Sham (n = 5), UIRI + Veh (n = 6), UIRI + ACEI (n = 6), and UIRI + Ncls (n = 6). (**d**) Representative Western blotting and quantification for fibronectin (FN), collagen I (COL1), and α-smooth muscle actin (α-SMA) at day 14 after UIRI. n = 6 in each group. (**e**) Representative images of hematoxylin and eosin (H&E), Masson trichrome, and Sirius Red staining of renal cortex at day 14 after UIRI, along with quantitative analysis of the fibrotic area and tubular injury score. n = 6 in each group. (**f**) Kidney mRNA expression of proinflammatory cytokines *Tnfα*, *Il6*, and *Il1b* determined by qPCR. n = 6 in each group. (**g**) Representative images of F4/80 immunohistochemistry staining in the UIRI mouse model, along with quantitative analysis of the positive area. n = 6 in each group. All data are presented as mean ± SD. $^{\#}P < 0.05$, $^{*}P < 0.01$. GAPDH, glyceraldehyde-3-phosphate dehydrogenase; IHC, immunohistochemistry; NS, not significant. To optimize viewing of this image, please see the online version of this article at www.kidney-international.org.

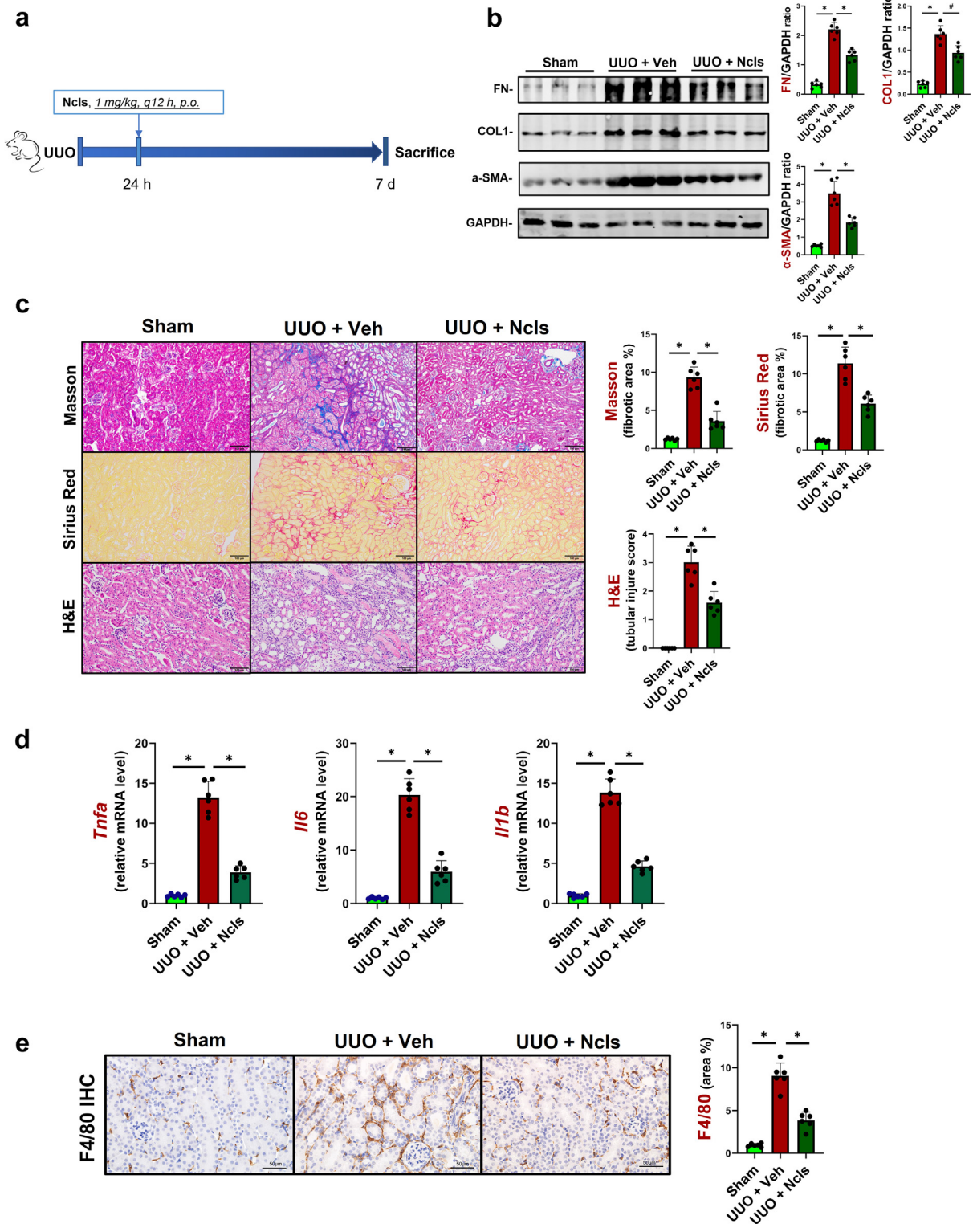


Figure 3 | Narciclasine (Ncls) alleviates tubulointerstitial fibrosis (TIF) and inflammation induced by unilateral ureteral obstruction (UUO). (**a**) Experimental design schematic in the UUO mouse model. (**b**) Representative Western blotting and quantification for fibronectin (FN), collagen I (COL1), and α-smooth muscle actin (α-SMA) at day 7 after UUO. (**c**) Representative images of hematoxylin and eosin (H&E), Masson trichrome, and Sirius Red staining of renal cortex at day 7 after UUO, along with quantitative analysis of the fibrotic area and tubular injury score. (**d**) Kidney mRNA expression of proinflammatory cytokines *Tnfα*, *Il6*, and *Il1b* determined by quantitative polymerase chain reaction. (**e**) Representative images of F4/80 immunohistochemistry staining, along with quantitative analysis of the positive area. All data are presented as mean ± SD. $^{\#}P < 0.05$, $^{*}P < 0.01$, versus UUO + vehicle (Veh) group, n = 6 in each group. GAPDH, glyceraldehyde-3-phosphate dehydrogenase. To optimize viewing of this image, please see the online version of this article at www.kidney-international.org.

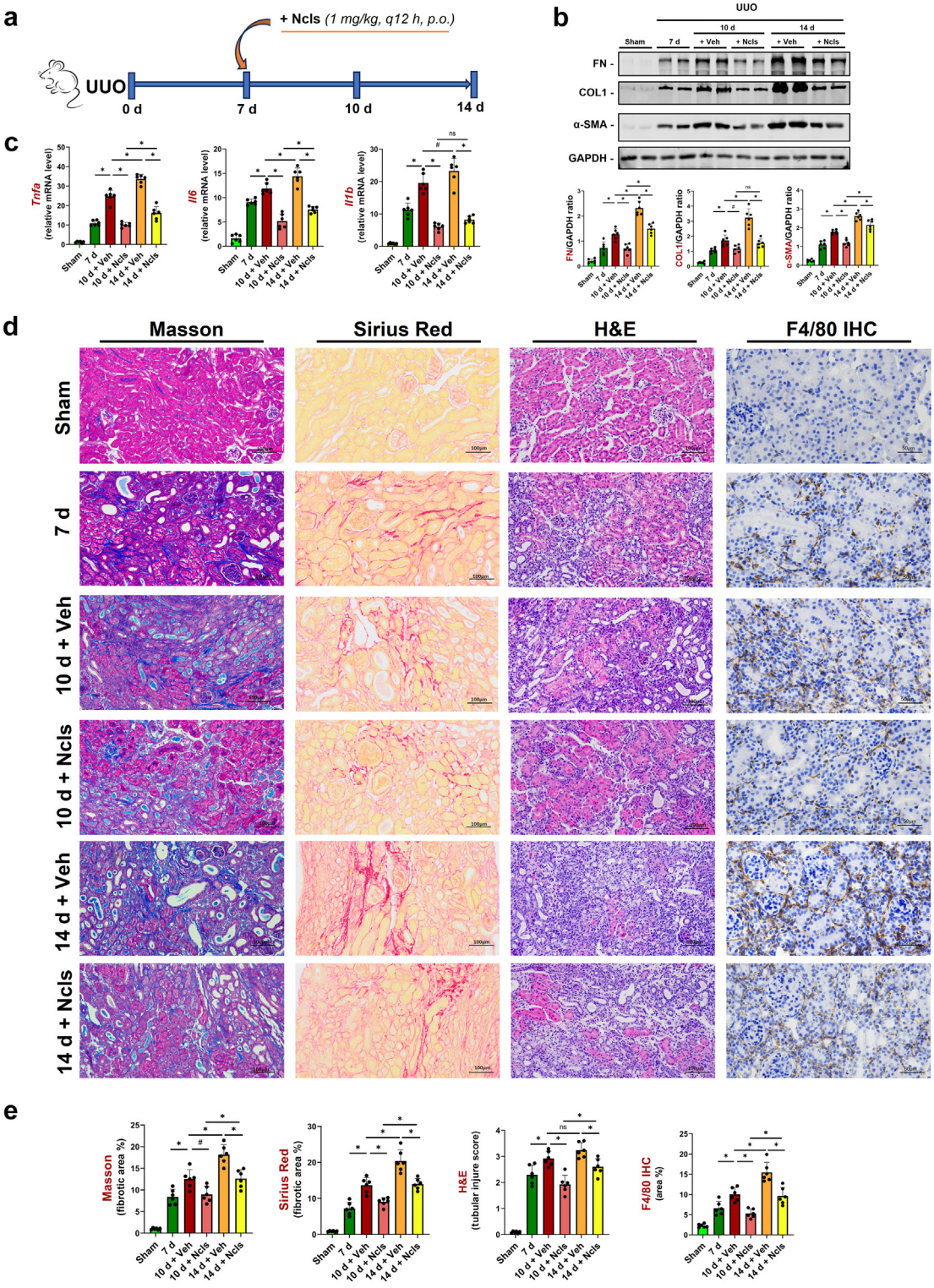


Figure 4 | Narciclasine (Ncls) mitigates the progression of established tubulointerstitial fibrosis (TIF) in the unilateral ureteral obstruction (UUO) mouse. (**a**) Experimental design schematic in the UUO mouse model. (**b**) Representative Western blotting and quantification for fibronectin (FN), collagen I (COL1), and α-smooth muscle actin (α-SMA) in kidney tissues at days 7, 10, and 14 after UUO. (**c**) Kidney mRNA expression levels of proinflammatory cytokines *Tnfα*, *Il6*, and *Il1b* were assessed in kidney tissues at days 7, 10, and 14 after UUO. (**d**) Representative histologic images of kidney sections at days 7, 10, and 14 after UUO stained with Masson's trichrome, Sirius Red, and hematoxylin and eosin (H&E). F4/80 immunohistochemical staining was used to visualize macrophage infiltration. (**e**) Quantitative analysis of Masson's trichrome, Sirius Red, H&E staining, and F4/80 immunohistochemistry (IHC). All data are presented as mean ± SD. $^{\#}P < 0.05$, $^{*}P < 0.01$, n = 6 in each group. GAPDH, glyceraldehyde-3-phosphate dehydrogenase; Veh, vehicle. To optimize viewing of this image, please see the online version of this article at www.kidney-international.org.

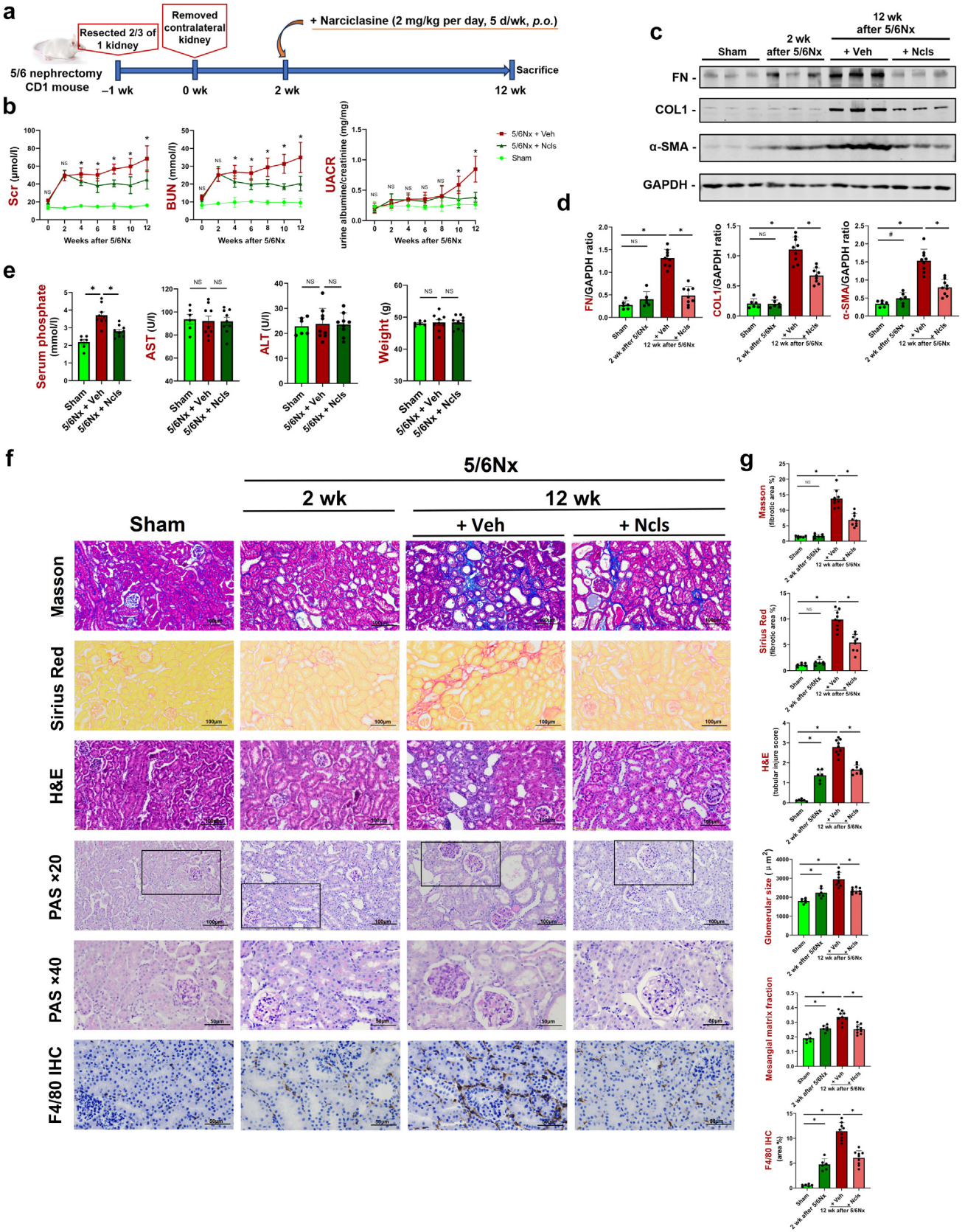


Figure 5 | Narciclasine (Ncls) mitigates the progression of tubulointerstitial fibrosis (TIF) in the 5/6 nephrectomy (5/6Nx) mouse model. (**a**) Experimental design timeline for the 5/6Nx CD-1 mouse model. Ncls treatment was initiated 2 weeks after surgery. (**b**) The line graphs depicting the changes in blood urea nitrogen (BUN), serum creatinine (Scr), and urinary albumin/creatinine (continued)

BRD-K04853698 (also known as LDN193189), 15-delta-prostaglandin-j2 (15d-PGJ_2), niclosamide, and trichostatin A, supporting the reliability of our analytical methodology.[14,26–31] However, significant challenges persist in translating these compounds into clinical practice.[26–35] Ncls, along with BRD-K92317137 and BRD-K04853698, consistently ranked among the top 3 compounds in all 7 datasets (Figure 1g). We selected Ncls for further experimental investigation based on its ability to inhibit nuclear factor κB (NF-κB) and signal transducer and activator of transcription 3 signaling pathways, indicating its potential role in combating TIF.[36–38] The genes reversed by Ncls in each dataset are listed in Supplementary Table S4.

Narciclasine alleviates TIF and renal inflammation induced by UIRI and UUO

We used a UIRI-induced CKD model to assess the antifibrotic effect of Ncls. Lisinopril, a renin-angiotensin system inhibitor (angiotensin-converting enzyme inhibitor), was used as a positive control. Oral administration of Ncls/lisinopril was initiated 3 days after UIRI surgery, and the mice were sacrificed on day 14 (Figure 2a). No significant changes were observed in body weight or liver function after Ncls treatment (Figure 2c). Both Ncls and lisinopril significantly reduced the level of serum creatinine and blood urea nitrogen, with Ncls showing superior efficacy (Figure 2b).

Western blot analysis revealed that Ncls markedly decreased the levels of fibronectin (FN), collagen I (COL1), and α-smooth muscle actin (α-SMA) in fibrotic renal lesions, exceeding the effects of lisinopril (Figure 2d). Hematoxylin and eosin staining revealed a significant reduction in the extent of renal tubular injury with Ncls treatment (Figure 2e). Masson's trichrome and Sirius Red staining showed a significant reduction in ECM accumulation after Ncls treatment (Figure 2e). Real-time polymerase chain reaction showed that Ncls significantly downregulated the gene expression of key inflammatory cytokines, including *Tnfa*, *Il6*, and *Il1b* (Figure 2f). Immunohistochemical staining demonstrated a significant reduction in F4/80$^+$ macrophage infiltration in the kidneys (Figure 2g). Taken together, these data indicate that Ncls attenuates UIRI-induced TIF and tubule-interstitial inflammation.

To validate the above findings, we used the UUO model (Figure 3a). Ncls was administered 24 hours after operation, and the mice were sacrificed on day 7. Western blot analysis revealed that Ncls markedly decreased the levels of FN, COL1, and α-SMA (Figure 3b). Masson's trichrome and Sirius Red staining confirmed that Ncls significantly reduced the deposition of collagen fibers (Figure 3c). Hematoxylin and eosin staining demonstrated that Ncls significantly ameliorated renal tubular injury induced by UUO (Figure 3c). Besides, Ncls significantly downregulated the expression of *Tnfa*, *Il6*, and *Il1b* (Figure 3d). Immunohistochemical staining demonstrated a significant reduction in F4/80$^+$ macrophage infiltration in the kidneys of UUO mice treated with Ncls (Figure 3e). In conclusion, our findings suggest that Ncls effectively mitigates renal inflammation and TIF induced by UUO.

To further assess the therapeutic effects of Ncls on established TIF, we initiated Ncls treatment on day 7 after UUO surgery. Western blotting, Masson's trichrome staining, and Sirius Red staining showed the presence of evident TIF on day 7 (Figure 4b and d). To evaluate the impact of Ncls on the progression of TIF, 6 mice from both the vehicle and Ncls-treated groups were randomly selected for assessment on days 10 and 14 after UUO. Histologic analysis using hematoxylin and eosin, Masson, and Sirius Red staining, along with immunohistochemical staining, revealed a consistent increase in tubular damage, F4/80$^+$ macrophage infiltration, and ECM deposition in the untreated group from day 7 to day 14 (Figure 4d and e). Real-time polymerase chain reaction analysis demonstrated a continuous upregulation of *Tnfa*, *Il6*, and *Il1b* (Figure 4c). Western blot analysis showed a progressive increase in the protein levels of FN, COL1, and α-SMA during the same period (Figure 4b). In the Ncls-treated group, although inflammation and TIF continued to worsen from day 7 to day 14, the severity of TIF and inflammation on days 10 and 14 after UUO was significantly reduced compared with the untreated group (Figure 4c–e). These findings indicate that Ncls effectively delays the progression of established TIF.

Narciclasine mitigates the progression of TIF induced by the 5/6 nephrectomy-induced CKD model

The 5/6 nephrectomy (5/6Nx) model is a classic animal model for CKD. As the disease progresses, 5/6Nx mice exhibit progressive renal dysfunction and significant proteinuria, along with histopathologic changes such as glomerular lesion and interstitial fibrosis.[16–18] Therefore, we further evaluated the efficacy of Ncls in the 5/6Nx model.

Ncls treatment was initiated 2 weeks after surgery, at which point the mice displayed significant elevations in serum

Figure 5 | (continued) ratio (UACR). *$P < 0.01$, 5/6Nx + Ncls versus 5/6Nx + vehicle (Veh) group. (**c**) Representative Western blotting images of fibronectin (FN), collagen I (COL1), and α-smooth muscle actin (α-SMA) expression in kidney tissues. (**d**) Quantification of Western blot data from panel (**c**). (**e**) Statistical analysis of serum phosphate levels, aspartate aminotransferase (AST), alanine aminotransferase (ALT), and body weight at 12 weeks after surgery. (**f**) Representative histologic images at 2 weeks after surgery in the 5/6Nx group and at 12 weeks after surgery in the Sham, 5/6Nx + Veh, and 5/6Nx + Ncls groups. Staining includes Masson's trichrome, Sirius Red, hematoxylin and eosin (H&E), periodic acid–Schiff (PAS), and immunohistochemical staining for F4/80. (**g**) Quantitative analysis of histologic staining (Masson's trichrome, Sirius Red, H&E, PAS, and F4/80 immunohistochemistry [IHC]) from panel (**f**). All data are presented as mean ± SD. Statistical significance: #$P < 0.05$, *$P < 0.01$. Sham (n = 6), 2 weeks after 5/6Nx (n = 6), 12 weeks after 5/6Nx + Veh (n = 9), 12 weeks after 5/6Nx + Ncls (n = 9). GAPDH, glyceraldehyde-3-phosphate dehydrogenase; NS, not significant; UUO, unilateral ureteral obstruction. To optimize viewing of this image, please see the online version of this article at www.kidney-international.org.

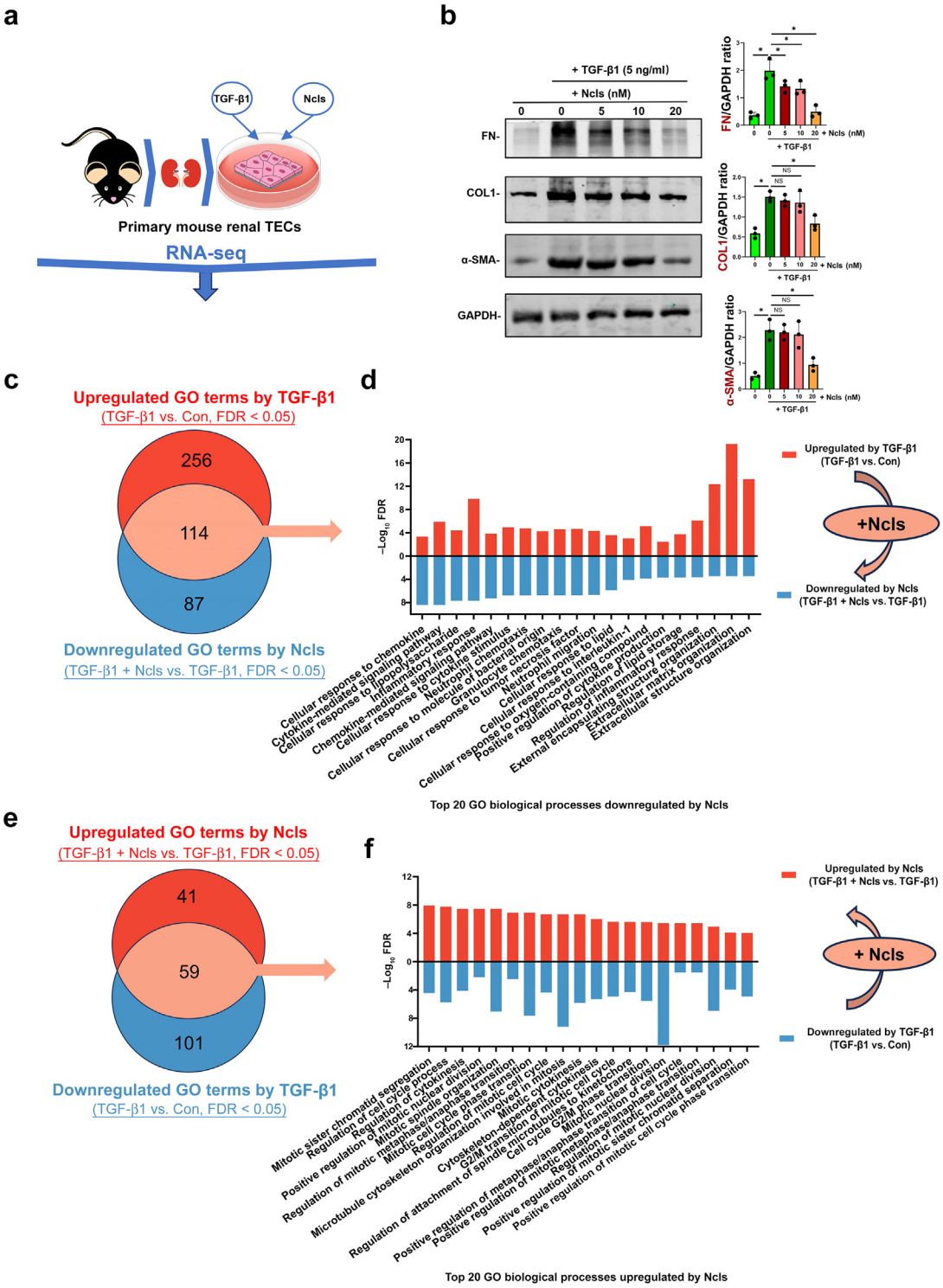


Figure 6 | Narciclasine (Ncls) reverses the transforming growth factor-β1 (TGF-β1)–induced biological processes in primary mouse renal tubular epithelial cells (TECs). (**a**) Schematic diagram of the *in vitro* experiments conducted on primary mouse TECs. (**b**) Representative Western blotting images and the quantification data showed the expression of fibronectin (FN), collagen I (COL1), and α-smooth muscle actin (α-SMA) in primary mouse TECs treated with TGF-β1 and indicated concentrations of Ncls. Data are presented as mean ± SD. $^{\#}P < 0.05$, $^{*}P < 0.01$, versus TGF-β1–treated cells, n = 3 in each group. (**c**) The Venn diagram shows the overlapping terms of Gene Ontology (GO) biological processes activated by TGF-β1 and inhibited by Ncls (20 nM). (**d**) The top 20 GO biological processes that are significantly inhibited by Ncls. The X-axis corresponds to the false discovery rate (FDR), arranged in ascending order, for the comparison between the TGF-β1 + Ncls group and the TGF-β1 group. (**e**) The Venn diagram shows the overlapping terms of GO biological processes inhibited by TGF-β1 and activated by Ncls (20 nM). (**f**) The top 20 GO biological processes significantly activated by Ncls. The X-axis corresponds to the FDR, arranged in ascending order, for the comparison between the TGF-β1 + Ncls group and the TGF-β1 group. GAPDH, glyceraldehyde-3-phosphate dehydrogenase; NS, not significant.

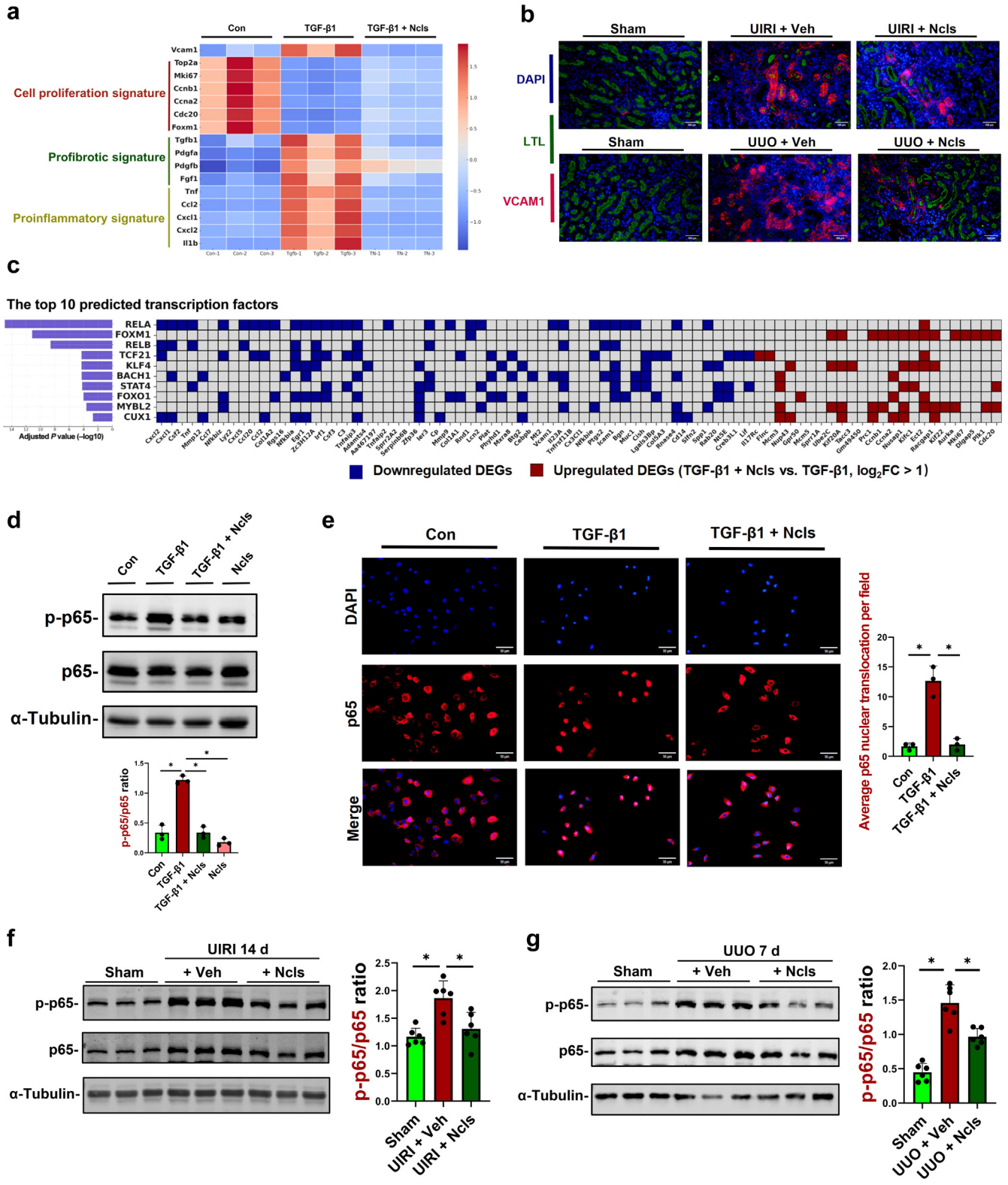


Figure 7 | Narciclasine (Ncls) inhibits transforming growth factor-β1 (TGF-β1)–induced failed repaired tubular epithelial cell (FR-TEC) phenotype via the nuclear factor κB signaling pathway. (**a**) Heatmap illustrating the expression of characteristic genes associated with FR-TECs in mouse primary TECs stimulated by TGF-β1 and Ncls. (**b**) Immunofluorescence analysis of vascular cell adhesion molecule 1 (VCAM1)–positive proximal tubular cells in unilateral ischemia-reperfusion injury (UIRI) and unilateral ureteral obstruction (UUO) mouse models. The figure presents the immunofluorescence staining of VCAM1-positive cells (indicated by red fluorescence) in comparison to healthy proximal tubular cells (marked by Lotus tetragonolobus lectin [LTL] represented in green fluorescence) in the kidneys of mice subjected to both UIRI and UUO models. The nuclei of these cells are visualized via 4′,6-diamidino-2-phenylindole (DAPI) staining (blue fluorescence). (**c**) The top 10 (continued)

creatinine and blood urea nitrogen levels (Figure 5a and b), and histologic changes such as renal tubular damage, glomerular hypertrophy, and mesangial matrix expansion, but without obvious collagen deposition in the renal interstitium (Figure 5f and g). Western blotting revealed a modest increase in α-SMA expression (Figure 5c and d). By the 12 weeks after surgery, the levels of serum creatinine, blood urea nitrogen, and urine albumin were dramatically increased (Figure 5b). Histopathologic analysis showed extensive tubular damage, interstitial fibrosis, and inflammatory cell infiltration. In addition, glomerular hypertrophy and mesangial matrix expansion were prominently observed (Figure 5f and g). In contrast, Ncls treatment notably mitigated these pathologic alterations (Figure 5f and g). Western blotting analysis confirmed that Ncls treatment significantly reduced the expression of FN, COL1, and α-SMA (Figure 5c and d). Furthermore, a significant increase in serum phosphate levels was noted in the 5/6Nx mice, which was notably reduced by Ncls treatment (Figure 5e).

Narciclasine reverses TGF-β1–induced biological processes in primary mouse renal tubular epithelial cells

The response to injury and subsequent transformation of tubular epithelial cells (TECs) play a pivotal role in the progression of TIF.[39] To assess the effects of Ncls on the profibrotic response of TECs, primary mouse TECs were stimulated with transforming growth factor-β1 (TGF-β1). Western blotting showed that Ncls dramatically decreased the expression of FN, COL1, and α-SMA in a dose-dependent manner. The most pronounced reduction was observed a concentration of 20 nM (Figure 6b).

To further explore the molecular mechanisms underlying these effects, we treated primary mouse TECs with TGF-β1 (5 ng/ml) and Ncls (20 nM) for 24 hours and then conducted RNA sequencing analysis. In the comparison between the TGF-β1 + Ncls cotreatment group and the TGF-β1 alone group, a total of 24 genes were significantly upregulated by Ncls ($\log_2$FC >1 and *adj.* $P < 0.05$), including well-known cell proliferation markers such as *Mki67*, *Ccnb1*, and *Cdc20* (Supplementary Table S5). In addition, 58 genes were significantly downregulated by Ncls ($\log_2$FC < −1 and *adj.* $P < 0.05$), which included classic profibrotic genes such as *Col1a1*, *Col1a2*, *Mmp9*, and *Mmp12*, as well as inflammatory factors such as *Tnf*, *Ccl2*, *Cxcl2*, and *Lif* (Supplementary Table S5). GO enrichment analysis revealed that TGF-β1 upregulated 370 categories, of which 114 were significantly suppressed by Ncls (Figure 6c). The top 20 Ncls-inhibited terms, ranked by statistical significance, were predominantly related to inflammation and ECM organization (Figure 6d). Moreover, the inhibitory effect of TGF-β1 on the proliferation of TECs was reversed by Ncls (Figure 6e and f).

Narciclasine inhibits TGF-β1–induced failed repaired TEC phenotype via the NF-κB signaling pathway

Research indicates that failed repair TECs (FR-TECs) are key contributors to maladaptive repair, driving the progression from acute kidney injury to CKD.[40–44] GO enrichment analysis revealed that TGF-β1 induces FR-TEC phenotype in primary mouse TECs (Figure 7a). Immunofluorescence staining revealed upregulated vascular cell adhesion molecule 1 (VCAM1), a marker of FR-TEC,[41–44] in TECs of both UIRI and UUO models (Figure 7b). Ncls significantly decreased VCAM1-positive TECs in both UIRI and UUO models (Figure 7b). Moreover, phenotype with FR-TECs was significantly reversed by Ncls (Figure 7a).

To decipher the mechanisms by which Ncls reverses TGF-β1–induced FR-TEC phenotypes, we conducted a transcription factor analysis on DEGs. RELA (p65) was identified as the top-ranked transcription factor regulating the majority of DEGs downregulated by Ncls (Figure 7c). Consistently, previous studies have shown the activation of NF-κB signaling in FR-TECs, and the p65 regulon is essential in the formation of FR-TECs.[40–42] The downregulated DEGs regulated by RELA includes proinflammatory factors (*Tnf*, *Ccl2*, and *Cxcl1*), profibrotic factors (*Col1a2*, *Mmp12*, and *Mmp9*), the FR-PT marker *Vcam1*, and the renal tubular cell injury marker *Lcn2* (Figure 7c). In support, Ncls significantly inhibits TGF-β1–induced phosphorylation of p65 (p-p65) in primary mouse TECs (Figures 7d). Immunofluorescence staining confirmed that Ncls interfered TGF-β1–induced nuclear translocation of p65 (Figures 7e). In addition, Ncls markedly reduced the level of p-p65 in renal fibrotic lesions in both the UIRI and UUO mice models (Figure 7f and g). Collectively, these findings suggest that Ncls inhibits the formation of the FR-TEC phenotype, at least in part, by suppressing the NF-κB signaling pathway.

Besides, we observed that most DEGs upregulated by Ncls, including *Mki67*, *Ccnb1*, and *Cdc20*, which regulate cell proliferation and differentiation, were the target genes of the transcription factor FOXM1 (Figure 7c), which is a key transcription factor for the effective differentiation and repair of dedifferentiated renal tubular cells after injury.[45] Notably,

Figure 7 | (continued) predicted transcription factors with the most statistically significant associations, as determined by the CHEA database, based on differentially expressed genes (DEGs; fold change [FC] >2) between the TGF-β1 + Ncls and TGF-β1 groups, along with their respective target genes. The X-axis represents DEGs (FC < −2 or >2) between the TGF-β1 + Ncls and TGF-β1 groups, sorted from left to right by increasing FC values. The blue blocks represent genes with an FC less than −2, whereas the red blocks represent genes with an FC greater than 2. (**d**) Western blot assessing the level of total p65, phosphorylated p65 (p-p65) in mouse primary TECs stimulated with Ncls and TGF-β1 for 6 hours. Data are presented as mean ± SD, *$P < 0.01$, versus TGF-β1, n = 3 in each group. (**e**) Immunofluorescence staining to observe the translocation of p-p65 into the nucleus in mouse primary TECs, *$P < 0.01$, versus TGF-β1, n = 3 in each group. (**f,g**) Western blot analysis of p65, p-p65 in UIRI and UUO (at day 7) mouse kidneys, *$P < 0.01$, versus UIRI + vehicle (Veh)/UUO + Veh, n = 6 in each group. All data are presented as mean ± SD. Con, control. To optimize viewing of this image, please see the online version of this article at www.kidney-international.org.

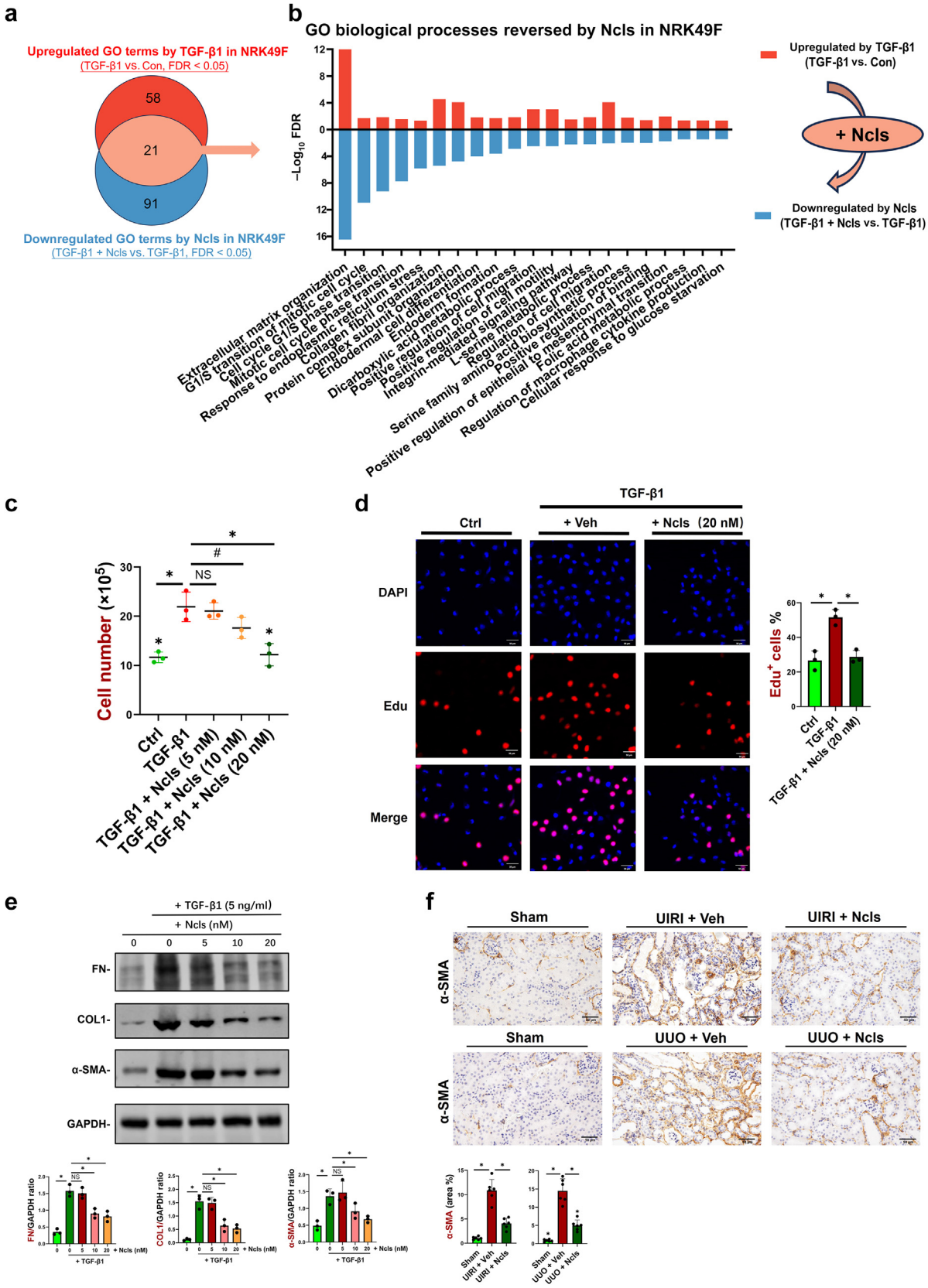


Figure 8 | Narciclasine (NcIs) blocks transforming growth factor-β1 (TGF-β1)–induced fibroblast proliferation and activation. (**a**) The Venn diagram shows the overlapping terms of Gene Ontology (GO) biological processes activated by TGF-β1 and inhibited by NcIs (20 nM) in rat renal fibroblast cells (NRK49F cells). (**b**) The GO biological processes that are significantly inhibited by NcIs in NRK49F cells. The X-axis corresponds to the FDR, arranged in ascending order, for the comparison between the TGF-β1 + NcIs group and the TGF-β1 (continued)

FOXM1 expression was also significantly upregulated by Ncls (Figure 7a), which might contribute to Ncls-promoted TEC proliferation and differentiation.

Narciclasine blocks TGF-β1–induced fibroblast proliferation and activation

Given the pivotal role of fibroblast activation in fibrotic scar formation, we further treated rat renal fibroblasts (NRK49F cells) with TGF-β1 (5 ng/ml) and Ncls (20 nM) for 24 hours, followed by RNA sequencing analysis. In the comparison between the TGF-β1 + Ncls cotreatment group and the TGF-β1 alone group, a total of 486 genes were significantly upregulated by Ncls ($log_2FC > 1$ and *adj.* $P < 0.05$). These genes were primarily enriched in biological processes related to ribosome biogenesis and protein translation (Supplementary Table S6). In addition, 963 genes were significantly downregulated by Ncls ($log_2FC < -1$ and *adj.* $P < 0.05$), with enrichment in processes related to ECM formation and cell proliferation (Supplementary Table S6). GO enrichment analysis also revealed that TGF-β1 activated biological processes associated with ECM accumulation and cell proliferation (Figure 8a and b), which was significantly inhibited by Ncls (Figure 8b). This observation was further validated experimentally. Cell count and 5-ethynyl-2′-deoxyuridine (EdU) staining consistently demonstrated a significant inhibitory effect of Ncls on TGF-β1–induced NRK49F cell proliferation (Figure 8c and d). Western blot analysis revealed a marked reduction in the expression of FN, COL1, and α-SMA (Figure 8e), which was corroborated with a notable decrease in α-SMA-positive renal stromal cells in the UUO mice treated with Ncls (Figure 8f). Collectively, these findings indicate that Ncls suppresses fibroblasts proliferation and activation.

DISCUSSION

TIF is the common feature of progressive CKD.[6] However, effective therapeutic strategies remain limited. The objective of the current study was to identify potential drug candidates for TIF. To this end, we focused on transcriptomic datasets derived from microdissected renal interstitial tissues of patients with CKD to construct the TIF GeneSig. This screening strategy offers several advantages. First, using data from patients with CKD rather than animal models ensures greater clinical relevance. Second, by leveraging transcriptomic data specifically from the renal tubular interstitium, we developed a GeneSig tailored to TIF, enabling the identification of candidate drugs with improved antifibrotic specificity. To assess the validity of the signature genes in reflecting the pathophysiological features of TIF, we performed both Reactome enrichment analysis and GO analysis, which confirmed the significant enrichment of fibrosis-related pathways in each GeneSig group. Finally, to enhance the reliability of our screening strategy, we evaluated the frequency of each compound identified across the selected datasets. Ncls was selected for further investigation, as it demonstrated the ability to effectively reverse the GeneSigs across the 7 selected datasets.

Ncls was initially recognized for its antitumor properties.[46] Recently, it has garnered attention for its anti-inflammatory effects, including peritonitis, sepsis, and acute lung injury.[37,46–48] Fuchs *et al.*[36] demonstrated the anti-inflammatory effects of Ncls on day 3 after UUO operation, corresponding to the acute phase of UUO-induced renal injury. However, whether Ncls exerts antifibrotic effect in any organs has not been explored. In the current study, first of all, we established that Ncls not only prevents TIF induced by UUO and UIRI, but also delays the progression of established TIF in UUO. Furthermore, we extended our observation to the 5/6 nephrectomy model, which closely mimic the pathophysiological features of human CKD. Our results demonstrate that Ncls effectively improved renal function, mitigated proteinuria, and alleviated TIF progression in the 5/6 nephrectomy model. Ncls also mitigated hyperphosphatemia in the 5/6 nephrectomy mouse model. These findings suggest its potential as a therapeutic candidate for CKD.

Recent studies indicate that FR-TECs are the key contributor to maladaptive repair, driving the progression from acute kidney injury to CKD.[40–44] Our study suggests that the phenotypic alterations in TECs induced by TGF-β1 align with the characteristics of FR-TECs. Notably, Ncls effectively counteracts FR-TEC phenotypes induced by TGF-β1. *In vivo*, Ncls reduced the number of VCAM1-positive FR-TECs. Single-cell RNA sequencing revealed that the hyperactivation of the p65 regulator was a distinct characteristic of FR-TECs.[40–42] We found that the genes upregulated by TGF-β1 and reversed by Ncls are primarily regulated by NF-κB. *In vitro*, Ncls blockaded TGF-β1–induced phosphorylation of p65 and its nuclear translocation in renal tubular cells. Collectively, inhibition of NF-κB signaling may contribute, at least in part, to the effect of Ncls on blocking the formation of the FR-TEC phenotype.

Despite the significant findings of our study, several limitations warrant consideration. First, the majority of drug gene

Figure 8 | (continued) group. (**c**) Quantification of NRK49F cell proliferation after 24 hours of exposure to TGF-β1 and indicated concentrations of Ncls. Data are presented as mean ± SD. $^{\#}P < 0.05$, $^{*}P < 0.01$, versus TGF-β1 group, n = 3 in each group. (**d**) Representation and quantitative analysis of NRK49F cell proliferation via EdU staining after 24 hours of treatment with TGF-β1 and Ncls, $^{*}P < 0.01$, versus TGF-β1 group, n = 3 in each group. (**e**) Representative Western blotting images, along with their respective quantification for fibronectin (FN), collagen I (COL1), and α-smooth muscle actin (α-SMA), are displayed. These measurements were derived from NRK49F cells treated with TGF-β1 and indicated concentrations of Ncls, $^{\#}P < 0.05$, $^{*}P < 0.01$, versus TGF-β1 group, n = 3 in each group. (**f**) Representative immunohistochemistry images of α-SMA in unilateral ischemia-reperfusion injury (UIRI) and unilateral ureteral obstruction (UUO; at day 7) mouse models, along with their respective quantitative analysis of the positive area. Data are presented as mean ± SD. $^{\#}P < 0.05$, $^{*}P < 0.01$, versus UIRI/UUO + vehicle (Veh) group, n = 6 in each group. Con, control; EdU, 5-ethynyl-2′-deoxyuridine; GAPDH, glyceraldehyde-3-phosphate dehydrogenase. To optimize viewing of this image, please see the online version of this article at www.kidney-international.org.

expression data in the LINCS database are derived from tumor cells, leading to a scarcity of renal-specific data. Nonetheless, the biological similarities between tumors and CKD, such as inflammatory microenvironments, epithelial-to-mesenchymal transition, and fibroblast activation, may account for the robustness of our screening results.[49,50] Second, Ncls, as an unmodified natural compound, exhibits pharmacokinetic limitations.[51] Future studies should aim to optimize its bioavailability and prolong its half-life. Third, mechanisms underlying the antifibrotic effect of Ncls have not been fully elucidated. Future investigations are required to identify the direct targets of Ncls.

In conclusion, we constructed a TIF GeneSig of patients with CKD and successfully identified Ncls as a potential agent to reverse the TIF GeneSig. We experimentally validated the antifibrotic effects of Ncls in several preclinical animal models including UUO, UIRI, and 5/6 nephrectomy mice. Collectively, our findings suggest that Ncls is a promising drug candidate for CKD treatment.

DISCLOSURE

All the authors declared no competing interests.

DATA STATEMENT

This study adheres to the data sharing policies of *Kidney International*. The RNA-Seq data supporting the findings of this study are openly available in the Gene Expression Omnibus at https://www.ncbi.nlm.nih.gov/geo/ under accession number GSE269356. Furthermore, the list of differentially expressed genes derived from the RNA-Seq data and the results of the enrichment analysis are provided in Supplementary Tables S5 and S6. The raw data used in the drug screening, including the list of differentially expressed genes from each human chronic kidney disease dataset, the enrichment analysis results, and the drug prediction outcomes based on the L1000CDS2 platform, are available in Supplementary Tables S1–S4 for reference. Additional data supporting the conclusions of this research are available from the corresponding author, Jing Nie, on reasonable request. Access to these datasets is open without restriction.

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AUTHOR CONTRIBUTIONS

JN supervised the project. JN and AX conceived and designed the project, and wrote the manuscript. AX completed transcriptome data analysis, statistical analysis of experimental data, and graphical drawing. AX, XC, and XMC worked together on *in vivo* and *in vitro* animal testing. ZH provided technique support. All the above work has been done with the crucial help of JM, TL, YM, QC, ZS, LZ, FZ. All authors read and reviewed the manuscript.

Supplementary material is available online at www.kidney-international.org.

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